UE24CS352A – Machine Learning Mini-Project

Application of Artificial Neural Networks in Streamflow Forecasting

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GitHub Repository Link :
https://github.com/sohel832/streamflow-forecasting-ann

1. Problem Statement

Streamflow forecasting is important for flood warning and water-resources planning, but the rainfall–runoff process is highly nonlinear and hard to model with physical or simple empirical formulas. The aim of this project is to predict the daily streamflow (a continuous value, so a regression problem) of the Leaf River near Collins, Mississippi, using only past rainfall and past streamflow. Two neural network models are built and compared: a standard Feed-Forward Neural Network (FFNN) and a Long Short-Term Memory (LSTM) network. The work follows the reference study "Application of Artificial Neural Network in Streamflow Forecasting" (Xiao and You, Stanford), in which the current runoff is assumed to be a function $Q(t) = f\{R(t-1),\ldots,R(t-m), Q(t-1),\ldots,Q(t-n)\}$.

2. Dataset Details

- **Streamflow:** USGS daily mean discharge (ft^3/s), site 02472000, parameter 00060, downloaded through the USGS Water Services API.
- **Rainfall:** NOAA GHCN-Daily precipitation (PRCP), station USC00221852 (Collins, MS), converted from tenths of mm to mm.
- **Period:** 1 Jan 2003 – 31 Dec 2012 (3,652 days). After removing 45 days with missing rainfall, 3,607 records remain.
- **Streamflow statistics:** mean 858 ft^3/s, median 280 ft^3/s, maximum 29,700 ft^3/s, showing a strongly skewed series with rare flood peaks.
- **Split:** chronological, with no shuffling. Training: 2003–2007; validation: 2008–2012 (about 1,800 samples each).

3. Approach

The two data series were merged by date, cleaned (negative and missing values removed) and converted into supervised-learning samples using lagged values. Features were scaled to the range 0–1 with Min-Max scaling, with the scalers fitted on the training data only to avoid data leakage. Both models were trained to minimise the Root Mean Square Error (RMSE), and predictions were converted back to the original scale before evaluation. Following the reference paper, errors are also reported separately for peak flow ($Q > 1500$) and non-peak flow ($Q \leq 1500$), and as a percentage of the maximum flow.

4. Implementation Overview

The project is implemented in Python (Jupyter notebook) using TensorFlow/Keras, pandas, NumPy, scikit-learn, matplotlib and seaborn. The notebook follows these steps: download raw data, load and clean, merge, create lag features, split and scale, build and train the models, predict, and evaluate. The README in the repository explains how to set up the environment and run the notebook.

	FFNN	LSTM
Input	7 lag features (3 rainfall, 4 streamflow)	Sequence of 3 days × 2 features (rainfall, streamflow)
Architecture	Dense(12, sigmoid) → Dense(1, linear)	LSTM(32, tanh; sigmoid gates) → Dense(1, linear)
Parameters	109	4,513
Optimizer / LR	Adam, 0.001	Adam, 0.002
Loss / metric	RMSE	RMSE
Training	200 epochs, batch 32, shuffle off, no early stopping	200 epochs, batch 32, shuffle off, no early stopping

Table 1. Model configurations

5. Results

Model	Validation RMSE (ft^3/s)	Peak RMSE (ft^3/s)	Non-peak RMSE (ft^3/s)	Training RMSE	Val. RMSE % of max
FFNN	733.52	2062.24	157.66	773.49	3.92%
LSTM	698.10	1961.20	152.67	635.33	3.73%

Table 2. Performance on the validation set (2008–2012; 218 peak days and 1,578 non-peak days)

The LSTM obtained a slightly lower validation RMSE than the FFNN, and both models were much more accurate on non-peak days than on peak days. Most of the total error comes from the 218 peak-flow days. The LSTM also has a lower training RMSE than validation RMSE (635 vs 698), which suggests mild overfitting, while the FFNN shows no such gap.

6. Conclusions

- Both neural networks can forecast daily streamflow using only past rainfall and streamflow, so they are practical alternatives to physical models when data is limited.
- In this implementation the LSTM (validation RMSE 698.10) performed slightly better than the FFNN (733.52). This differs from the reference paper, where the FFNN was better. The differences are likely due to our different choices of lag length, learning rate, scaling and data cleaning.
- Peak-flow prediction remains the main weakness: peak RMSE is about 13 times the non-peak RMSE for both models. Since underestimating floods is risky, a safety factor could be applied to predicted peaks.
- Future work: add more input features (e.g. temperature, soil and geographic data), tune the lag days and hidden sizes, and test the models on other river basins.

References

[1] Xiao, M. and You, S. Application of Artificial Neural Network in Streamflow Forecasting. Stanford University (reference report).

[2] Hsu, K., Gupta, H. V. and Sorooshian, S. (1995). Artificial neural network modeling of the rainfall-runoff process. Water Resources Research, 31(10), 2517–2530.

[3] Kratzert, F. et al. (2018). Rainfall–runoff modelling using long short-term memory (LSTM) networks. Hydrol. Earth Syst. Sci., 22(11), 6005–6022.

[4] Data: USGS Water Services (site 02472000); NOAA NCEI GHCN-Daily (station USC00221852).